9.7 Equal Sums of Powers

WORKSHEET

The question

STEP I 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q5

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If a polynomial $\mathrm{p}(x) = \sum c_k x^k$ has $\mathrm{p}(1) = 0$, what does that say about the coefficients? What about $\mathrm{p}'(1) = 0$?

STAGE 1 A triple root

Ask yourself Factorise $1 - x^2$ and $1 - x^4$. How many factors of $1 - x$ appear?

STAGE 2 Derivatives at 1

Ask yourself If $(1 - x)^3$ divides p_0, what power of $(1 - x)$ divides p_0'? And then $\mathrm{p}_1 = x\mathrm{p}_0'$ and p_1'?

STAGE 3 Reading off the coefficients

Ask yourself What are $p_0'(1)$ and $p_1'(1)$ in terms of the coefficients c_k?

STAGE 4 **One more factor**

Pause and try To make the cubes balance too, we need a fourth factor of $(1-x)$. Which polynomial should replace p_0?

What's really going on

The Thue–Morse sequence

The signs of the coefficients of $(1-x)(1-x^2)(1-x^4)(1-x^8)\cdots$ form the *Thue–Morse sequence*. Writing 0 for $+$ and 1 for $-$:

$$0\,1\,1\,0\,1\,0\,0\,1\,1\,0\,0\,1\,0\,1\,1\,0 \ldots$$

There are three equivalent descriptions.

- **Doubling:** start with 0, and repeatedly append the complement (swap 0s and 1s) of what you have: $0 \to 01 \to 0110 \to 01101001 \to \cdots$.
- **Binary digits:** the kth term is 0 if k has an even number of 1s when written in binary, and 1 if odd. (This is why the sign of x^k is $(-1)^{\text{number of 1s in } k}$: to make x^k from the product, you choose the factors x^{2^m} that make up k in binary, and each brings a minus sign.)
- **Polynomial:** the coefficients of $\prod_m (1-x^{2^m})$, as in the question.

Splitting $0, 1, \ldots, 2^N - 1$ by the Thue–Morse sequence gives two sets whose rth powers have equal sums for every $r < N$ (Figure 9.7.1). This is the *Prouhet–Tarry–Escott problem*:

find two different sets of integers with equal sums of rth powers for $r = 1, \dots, k$. Prouhet's solution uses 2^{k+1} numbers; it is a famous open problem to find out how few numbers are really needed.



$1 + 4 + 6 + 7 + 10 + 11 + 13 + 16 = 68$

$2 + 3 + 5 + 8 + 9 + 12 + 14 + 15 = 68$

squares: $748 = 748$ cubes: $9248 = 9248$ fourth powers: $122\,692 \neq 121\,156$

Figure 9.7.1 The numbers 1 to 16 split by the Thue–Morse sequence. The two sets have equal sums, equal sums of squares and equal sums of cubes, but not equal sums of fourth powers; for that you need 1 to 32. Reading the colours left to right gives the pattern teal–plum–plum–teal, plum–teal–teal–plum, ...

The fairest way to take turns

Two children take turns choosing sweets from a pile, best first. If they alternate $ABABAB\dots$, then A always gets the better sweet of each pair, a systematic advantage. The order $ABBA$ is fairer, and $ABBABAAB$ fairer still: this is the Thue–Morse sequence, and the balanced power sums are exactly the reason. If the sweets have values given by a smooth function of their position, expanding that function as a polynomial shows that the Thue–Morse order cancels its leading terms. The same idea has been proposed for penalty shoot-outs in football (the "$ABBA$" order trialled by football's rule-makers from 2017) and is used for the order of serving in tennis tie-breaks.

Thue, Morse and Euwe

The sequence has been discovered many times. Prouhet used it in 1851. Axel Thue rediscovered it in 1912 while studying sequences without repetitive patterns: it never contains the same block three times in a row (it is "cube-free"). Marston Morse found it again in 1921 in the study of geodesics on surfaces. And the chess world champion Max Euwe used it in 1929 to show that the rule then in force for drawing a game by repetition did not prevent infinitely long games. Today it is a central example in the theory of *automatic sequences*, sequences computed by simple machines reading the binary digits of n.

Try this next

1 **Shifting.** Try STEP II 2025, Question 8 (in OCR's download of the 2025 papers, www.ocr.org.uk/Images/750486-2025-past-papers.zip). It shows how to turn a split that balances sums and squares into a larger one that balances cubes as well, by shifting one list and combining.

2 **Squares of squares.** (Also from STEP II, 2025.) The sets $A = \{1, 3, 4, 5, 9, 11\}$ and $B = \{2, 6, 7, 8, 10\}$ have equal sums and equal sums of squares. Split $\{n^2, (n+1)^2, \ldots, (n+11)^2\}$ into two sets with equal sums, for every positive integer n. (*Hint:* add 0 to the second set.)

3 **Translation invariance.** Show that if two sets of numbers have equal sums of rth powers for $r = 0, 1, \ldots, k$, then so do the sets obtained by adding the same number c to every element. Why does this mean that the question's sets $\{1, \ldots, 15\}$ could equally be replaced by any 16 consecutive integers?

4 **Fair sweets.** Sweets have values $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{8}$. Compare the totals A gets with the orders $ABABABAB$ and $ABBABAAB$.

Further reading

- **Jean-Paul Allouche & Jeffrey Shallit, "The ubiquitous Prouhet–Thue–Morse sequence"**, in *Sequences and Their Applications* (Springer, 1999). A survey of the sequence's many appearances; free online.
- **Matt Parker, *Things to Make and Do in the Fourth Dimension*** (Penguin). Includes equal power sums and other number curiosities.
- **Joshua Cooper & Aaron Dutle, "Greedy Galois games"**, *American Mathematical Monthly* 120 (2013). Fair turn-taking and Thue–Morse.

The full guided solution is in Section 9.7:
https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html